# Lecture notes — Itô examples & first steps toward Girsanov

(Board-ready derivations; focus: explicit computations + intuition for change of measure)

## Part A — Itô formula: 40–45 minutes of explicit computations

### A.1 Warm-up: the prototype computation

Let $(W_t)_{t\geq 0}$ be a Brownian motion and recall Itô's formula: for $f \in C^{1,2}([0,T]\times\mathbb{R})$ and an Itô process $X_t$ with

$$\mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t,$$

we have

$$\mathrm{d}f(t,X_t) = \partial_t f(t,X_t)\ \mathrm{d}t + \partial_x f(t,X_t)\ \mathrm{d}X_t + \frac{1}{2}\partial_{xx} f(t,X_t)\,(\mathrm{d}X_t)^2, \qquad (\mathrm{d}X_t)^2 = \sigma_t^2\ \mathrm{d}t.$$

- Always separate: **(drift)** $\mathrm{d}t$ part vs **(martingale)** $\mathrm{d}W$ part.
- Remember: $(\mathrm{d}W_t)^2 = \mathrm{d}t,\ \mathrm{d}t\ \mathrm{d}W_t = 0,\ (\mathrm{d}t)^2 = 0.$

### A.2 Example 1: $W_t^2 - t$ is a martingale

Apply Itô to $f(x) = x^2$ with $X_t = W_t$. We have $f'(x) = 2x$, $f''(x) = 2$, and $\mathrm{d}W_t = 0\cdot\mathrm{d}t + 1\cdot\mathrm{d}W_t$:

$$\mathrm{d}(W_t^2) = 2W_t\ \mathrm{d}W_t + \frac{1}{2}\cdot 2\,(\mathrm{d}W_t)^2 = 2W_t\ \mathrm{d}W_t + \mathrm{d}t.$$

Hence

$$\mathrm{d}(W_t^2 - t) = 2W_t\ \mathrm{d}W_t,$$

so $W_t^2 - t$ is an Itô integral and therefore a martingale (integrable, mean 0 increment).

**What to highlight.** The *only* reason we need the "$-t$" is the Itô correction term.

### A.3 Example 2: Exponential martingale ("kill the drift")

Let $Y_t = \exp(at + bW_t)$, with constants $a, b \in \mathbb{R}$. Use Itô with $f(t,x) = e^{at+bx}$. Compute derivatives:

$$\partial_t f = af, \qquad \partial_x f = bf, \qquad \partial_{xx} f = b^2 f.$$

With $X_t = W_t$:

$$\mathrm{d}Y_t = aY_t\ \mathrm{d}t + bY_t\ \mathrm{d}W_t + \tfrac{1}{2}b^2 Y_t\ \mathrm{d}t = Y_t\Big((a + \tfrac{1}{2}b^2)\ \mathrm{d}t + b\ \mathrm{d}W_t\Big).$$

Choose $a = -\frac{1}{2}b^2$ to remove the drift:

$$\mathrm{d}Y_t = bY_t\ \mathrm{d}W_t, \quad \Rightarrow \quad Y_t = \exp\Big(bW_t - \tfrac{1}{2}b^2 t\Big) \text{ is a martingale.}$$

**Pedagogical link.** This is the prototype of the Radon–Nikodym density used in Girsanov.

### A.4 Example 3: Geometric Brownian motion and the log trick

Consider GBM:

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0.$$

Let $f(x) = \ln x$, $f'(x) = 1/x$, $f''(x) = -1/x^2$. Then

$$\mathrm{d}(\ln S_t) = \frac{1}{S_t}\,\mathrm{d}S_t + \frac{1}{2}\Big(-\frac{1}{S_t^2}\Big)(\,\mathrm{d}S_t)^2 = \frac{1}{S_t}(\mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t) - \frac{1}{2}\frac{1}{S_t^2}(\sigma^2 S_t^2\,\mathrm{d}t),$$

so

$$\mathrm{d}(\ln S_t) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Integrate:

$$\ln S_t = \ln S_0 + \Big(\mu - \tfrac{1}{2}\sigma^2\Big)t + \sigma W_t, \qquad S_t = S_0 \exp\Big(\Big(\mu - \tfrac{1}{2}\sigma^2\Big)t + \sigma W_t\Big).$$

**Board payoff.** You get the explicit solution + lognormal law in one shot.

### A.5 Example 4: Discounting trick (product rule, no PDE yet)

Let $A_t$ be a deterministic (or finite-variation adapted) process with

$$\mathrm{d}A_t = -\gamma_t A_t\,\mathrm{d}t, \qquad A_0 = 1,$$

and let $S_t$ satisfy $\mathrm{d}S_t = \mu_t S_t\,\mathrm{d}t + \sigma_t S_t\,\mathrm{d}W_t$. Define $X_t := A_t S_t$. Use product rule:

$$\mathrm{d}X_t = A_t\,\mathrm{d}S_t + S_t\,\mathrm{d}A_t + \mathrm{d}[A,S]_t.$$

Here $A$ has zero quadratic variation, so $\mathrm{d}[A,S]_t = 0$, hence

$$\mathrm{d}X_t = A_t(\mu_t S_t\,\mathrm{d}t + \sigma_t S_t\,\mathrm{d}W_t) - \gamma_t A_t S_t\,\mathrm{d}t = X_t(\mu_t - \gamma_t)\,\mathrm{d}t + X_t\sigma_t\,\mathrm{d}W_t.$$

Choose $\gamma_t = \mu_t$:

$$\mathrm{d}X_t = X_t\sigma_t\,\mathrm{d}W_t,$$

so $(X_t)$ is a martingale (under suitable integrability, e.g. $\mathbb{E}\int_0^T X_t^2\sigma_t^2\,\mathrm{d}t < \infty$).

**Message.** "Find the right discount factor ⇒ spot the right drift for pricing."

### A.6 (If time) Example 5: A simple Itô identity to remove a stochastic integral

Goal: compute an expression for $\int_0^t e^{W_s}\,\mathrm{d}W_s$ *without* stochastic integral.

Take $f(x) = e^x$ and $X_t = W_t$:

$$\mathrm{d}(e^{W_t}) = e^{W_t}\,\mathrm{d}W_t + \tfrac{1}{2}e^{W_t}\,\mathrm{d}t.$$

Integrate from 0 to $t$:

$$e^{W_t} - 1 = \int_0^t e^{W_s}\,\mathrm{d}W_s + \frac{1}{2}\int_0^t e^{W_s}\,\mathrm{d}s,$$

so

$$\boxed{\int_0^t e^{W_s}\,\mathrm{d}W_s = e^{W_t} - 1 - \frac{1}{2}\int_0^t e^{W_s}\,\mathrm{d}s.}$$

## A.7: Norm squared $\|W_t\|^2$ and drift identification

Assume for simplicity $\Sigma = I_d$ (independent coordinates). Let $R_t := \|W_t\|^2 = \sum_{i=1}^d (W_t^i)^2$. Using $\mathrm{d}(W^i)^2 = 2W^i\,\mathrm{d}W^i + \mathrm{d}t$ and summing:

$$\mathrm{d}R_t = 2\sum_{i=1}^d W_t^i\,\mathrm{d}W_t^i + d\,\mathrm{d}t = 2\,W_t^\top\,\mathrm{d}W_t + d\,\mathrm{d}t.$$

**Extract:** $\|W_t\|^2 - dt$ is a martingale. **Comment:** this shows the "dimension drift" $+d\,dt$.

## A.8: MD-8: Itô for $f(X_t^1, X_t^2)$ with correlated Brownian drivers

**Setup (correlated Brownian motion).** Let $(W_t^1, W_t^2)$ be a 2D Brownian motion with correlation $\rho \in [-1, 1]$:

$$\mathrm{d}[W^1]_t = \mathrm{d}t, \qquad \mathrm{d}[W^2]_t = \mathrm{d}t, \qquad \mathrm{d}[W^1, W^2]_t = \rho\,\mathrm{d}t.$$

Let $X^1, X^2$ be Itô processes driven by $(W^1, W^2)$:

$$\mathrm{d}X_t^1 = b_1(t)\,\mathrm{d}t + \sigma_{11}(t)\,\mathrm{d}W_t^1 + \sigma_{12}(t)\,\mathrm{d}W_t^2, \qquad \mathrm{d}X_t^2 = b_2(t)\,\mathrm{d}t + \sigma_{21}(t)\,\mathrm{d}W_t^1 + \sigma_{22}(t)\,\mathrm{d}W_t^2.$$

Let $f \in C^2(\mathbb{R}^2)$ and set $Y_t := f(X_t^1, X_t^2)$.

**Step 1: write the quadratic covariations.** Using bilinearity and $\mathrm{d}W^1\,\mathrm{d}W^2 = \rho\,\mathrm{d}t$:

$$\mathrm{d}[X^1]_t = \Big(\sigma_{11}^2 + \sigma_{12}^2 + 2\rho\,\sigma_{11}\sigma_{12}\Big)\,\mathrm{d}t,$$

$$\mathrm{d}[X^2]_t = \Big(\sigma_{21}^2 + \sigma_{22}^2 + 2\rho\,\sigma_{21}\sigma_{22}\Big)\,\mathrm{d}t,$$

$$\mathrm{d}[X^1, X^2]_t = \Big(\sigma_{11}\sigma_{21} + \sigma_{12}\sigma_{22} + \rho(\sigma_{11}\sigma_{22} + \sigma_{12}\sigma_{21})\Big)\,\mathrm{d}t.$$

**Step 2: apply 2D Itô formula.**

$$\mathrm{d}Y_t = f_{x_1}(X_t)\,\mathrm{d}X_t^1 + f_{x_2}(X_t)\,\mathrm{d}X_t^2 + \frac{1}{2}f_{x_1x_1}(X_t)\,\mathrm{d}[X^1]_t + f_{x_1x_2}(X_t)\,\mathrm{d}[X^1, X^2]_t + \frac{1}{2}f_{x_2x_2}(X_t)\,\mathrm{d}[X^2]_t.$$

**Board takeaway:** the correlation $\rho$ appears only through $\mathrm{d}[X^1]$, $\mathrm{d}[X^2]$, and especially $\mathrm{d}[X^1, X^2]$, which multiplies the mixed derivative $f_{x_1x_2}$.

### Concrete choice: $f(x_1, x_2) = x_1x_2$ (product)

Take $f(x_1, x_2) = x_1x_2$. Then

$$f_{x_1} = x_2, \quad f_{x_2} = x_1, \quad f_{x_1x_2} = 1, \quad f_{x_1x_1} = f_{x_2x_2} = 0.$$

Hence

$$\mathrm{d}(X_t^1X_t^2) = X_t^2\,\mathrm{d}X_t^1 + X_t^1\,\mathrm{d}X_t^2 + \mathrm{d}[X^1, X^2]_t.$$

Plug the formula for $\mathrm{d}[X^1, X^2]_t$ above to see explicitly where $\rho$ enters.

**Concrete choice:** $f(x_1, x_2) = \exp(\alpha x_1 + \beta x_2)$ **(drift-killing style)**

Let $f(x_1, x_2) = \exp(\alpha x_1 + \beta x_2)$, so

$$f_{x_1} = \alpha f, \quad f_{x_2} = \beta f, \quad f_{x_1 x_1} = \alpha^2 f, \quad f_{x_2 x_2} = \beta^2 f, \quad f_{x_1 x_2} = \alpha\beta f.$$

Then

$$\frac{\mathrm{d}Y_t}{Y_t} = \alpha \, \mathrm{d}X_t^1 + \beta \, \mathrm{d}X_t^2 + \frac{1}{2}\alpha^2 \, \mathrm{d}[X^1]_t + \alpha\beta \, \mathrm{d}[X^1, X^2]_t + \frac{1}{2}\beta^2 \, \mathrm{d}[X^2]_t.$$

**Board takeaway:** compared to 1D, the extra term is the cross-variation piece $\alpha\beta \, \mathrm{d}[X^1, X^2]_t$, which carries the correlation $\rho$.

# Part B — Black–Scholes hedging ⇒ PDE ⇒ expectation

## B.1 Model setup (GBM + bank account)

Assume a frictionless market with one risky asset $S$ and one money-market account $B$:

$$\frac{\mathrm{d}S_t}{S_t} = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}W_t, \qquad \mu \in \mathbb{R}, \ \sigma > 0,$$

$$\mathrm{d}B_t = rB_t \, \mathrm{d}t, \qquad B_0 = 1, \qquad B_t = e^{rt}.$$

Define the discount factor

$$D_t := B_t^{-1} = e^{-rt}.$$

- Parameters: $\mu$ (real drift), $\sigma$ (volatility), $r$ (short rate).
- $B_t = e^{rt}$ is deterministic; $D_t = e^{-rt}$ is its inverse.

## B.2 Derivative price as a Markov function and Itô expansion

Fix a maturity $T$ and a European payoff $g(S_T)$. Assume the time-$t$ price is given by a smooth deterministic function

$$V_t = p(t, S_t), \qquad 0 \le t \le T, \qquad p(T, s) = g(s).$$

By Itô's formula (with $(\mathrm{d}S_t)^2 = \sigma^2 S_t^2 \, \mathrm{d}t$),

$$\mathrm{d}p(t, S_t) = \Big(\partial_t p(t, S_t) + \mu S_t \, \partial_s p(t, S_t) + \tfrac{1}{2}\sigma^2 S_t^2 \, \partial_{ss} p(t, S_t)\Big) \, \mathrm{d}t + \sigma S_t \, \partial_s p(t, S_t) \, \mathrm{d}W_t.$$

- Write explicitly: *drift part + noise part.*
- The coefficient of $\mathrm{d}W_t$ is $\sigma S_t \, \partial_s p(t, S_t)$.

## B.3 Self-financing hedging portfolio and "kill the $\mathrm{d}W$"

Consider a self-financing portfolio holding $\Delta_t$ shares of stock and the rest in the bank account:

$$X_t := \Delta_t S_t + \beta_t B_t.$$

Self-financing means "no external cash injection":

$$\mathrm{d}X_t = \Delta_t \, \mathrm{d}S_t + \beta_t \, \mathrm{d}B_t.$$

Using $\beta_t B_t = X_t - \Delta_t S_t$ and $\mathrm{d}B_t = rB_t\ \mathrm{d}t$,

$$\mathrm{d}X_t = \Delta_t(\mu S_t\ \mathrm{d}t + \sigma S_t\ \mathrm{d}W_t) + r(X_t - \Delta_t S_t)\,\mathrm{d}t.$$

We aim to replicate the option: $X_t \equiv p(t, S_t)$, so $\mathrm{d}X_t = \mathrm{d}p(t, S_t)$. Comparing the $\mathrm{d}W_t$ **coefficients** gives the **delta hedge**:

$$\boxed{\Delta_t = \partial_s p(t, S_t)}.$$

- This is the "kill the noise" step: choose $\Delta_t$ to cancel the $\mathrm{d}W_t$ term.
- Once the noise is removed, the hedged position becomes locally riskless.

### B.4 Black–Scholes PDE (no-arbitrage)

Plug $\Delta_t = \partial_s p(t, S_t)$ into the drift matching. From the portfolio drift:

$$\mathrm{d}X_t = \Big(rX_t + (\mu - r)S_t\Delta_t\Big)\,\mathrm{d}t + \sigma S_t\Delta_t\ \mathrm{d}W_t.$$

From the Itô expansion:

$$\mathrm{d}p(t, S_t) = \Big(\partial_t p + \mu S_t\partial_s p + \tfrac{1}{2}\sigma^2 S_t^2\partial_{ss}p\Big)\,\mathrm{d}t + \sigma S_t\partial_s p\ \mathrm{d}W_t.$$

After setting $\Delta_t = \partial_s p(t, S_t)$, the $\mathrm{d}W_t$ terms match automatically. Matching the drift terms yields

$$r\,p(t, S_t) + (\mu - r)S_t\partial_s p(t, S_t) = \partial_t p(t, S_t) + \mu S_t\partial_s p(t, S_t) + \tfrac{1}{2}\sigma^2 S_t^2\partial_{ss}p(t, S_t).$$

The $\mu S_t\partial_s p$ cancels on both sides, giving the **valuation PDE**:

$$\boxed{\partial_t p(t, s) + rs\,\partial_s p(t, s) + \tfrac{1}{2}\sigma^2 s^2\,\partial_{ss}p(t, s) - r\,p(t, s) = 0, \qquad p(T, s) = g(s).}$$

- **Key point: $\mu$ disappears.** Prices do not depend on the real drift.
- This is the core Black–Scholes hedging argument.

## Solving the Black–Scholes PDE via the heat equation

We now solve

$$\partial_t p + rs\,\partial_s p + \tfrac{1}{2}\sigma^2 s^2\,\partial_{ss}p - rp = 0, \qquad p(T, s) = g(s).$$

### Step 1: time-to-maturity and discounting

Let

$$\tau := T - t, \qquad v(\tau, s) := p(T - \tau, s), \qquad v(0, s) = g(s).$$

Since $\partial_t p = -\partial_\tau v$, the PDE becomes

$$\partial_\tau v = rs\,\partial_s v + \tfrac{1}{2}\sigma^2 s^2\,\partial_{ss}v - rv.$$

Remove the $-rv$ term via discounting:

$$u(\tau, s) := e^{r\tau}v(\tau, s).$$

A short computation gives

$$\partial_\tau u = rs\,\partial_s u + \tfrac{1}{2}\sigma^2 s^2\,\partial_{ss}u, \qquad u(0, s) = g(s).$$

**Step 2: logarithmic change of variables**

Set

$$x := \ln s, \qquad w(\tau, x) := u(\tau, e^x).$$

Using

$$\partial_s u = \frac{1}{s} w_x, \qquad \partial_{ss} u = \frac{1}{s^2}(w_{xx} - w_x),$$

we obtain

$$\partial_\tau w = \tfrac{1}{2}\sigma^2 w_{xx} + \Big(r - \tfrac{1}{2}\sigma^2\Big) w_x, \qquad w(0, x) = g(e^x).$$

**Step 3: remove the transport term (shift)**

Let

$$a := r - \tfrac{1}{2}\sigma^2, \qquad \varphi(\tau, y) := w(\tau, y - a\tau).$$

Then $\varphi$ solves the heat equation

$$\partial_\tau \varphi = \tfrac{1}{2}\sigma^2\, \partial_{yy}\varphi, \qquad \varphi(0, y) = g(e^y).$$

**Step 4: heat equation solution ⇒ expectation form**

If $Z \sim \mathcal{N}(0, 1)$, the heat equation solution is

$$\varphi(\tau, y) = \mathbb{E}\Big[g\Big(e^{\,y + \sigma\sqrt{\tau}\, Z}\Big)\Big].$$

Undoing the changes of variables yields, for $\tau = T - t$,

$$\boxed{p(t, s) = e^{-r(T-t)}\mathbb{E}\Big[g\Big(s \exp\Big((r - \tfrac{1}{2}\sigma^2)(T - t) + \sigma\sqrt{T - t}\, Z\Big)\Big)\Big].}$$

*Remark* 0.1 (Pedagogical takeaway). The expectation representation comes from *solving the PDE* (via the heat equation). Next, Girsanov will explain how the same representation arises from a change of probability measure.

## Closed-form call formula and hedging

From now on, take $g(s) = (s - K)_+$ (European call with strike $K$). A direct computation of the expectation above yields the Black–Scholes formula.

**Theorem 0.2** (Black–Scholes call price)**.**

$$\boxed{p(0, S_0) = S_0\, \mathbf{N}(d_+) - K e^{-rT}\, \mathbf{N}(d_-),} \qquad d_\pm = \frac{\ln\big(\frac{S_0}{K e^{-rT}}\big)}{\sigma\sqrt{T}} \pm \frac{1}{2}\sigma\sqrt{T}.$$

**Theorem 0.3** (Delta hedge)**.**

$$\boxed{\Delta_t = \partial_s p(t, S_t) = \mathbf{N}(d_+(t, S_t)),} \qquad d_+(t, s) = \frac{\ln\big(\frac{s}{K e^{-r(T-t)}}\big)}{\sigma\sqrt{T - t}} + \frac{1}{2}\sigma\sqrt{T - t}.$$

### Greeks (definitions + interpretation)

Define the main sensitivities:

$$\Delta := \partial_s p, \qquad \Gamma := \partial_{ss} p, \qquad \Theta := \partial_t p, \qquad \text{Vega} := \partial_\sigma p, \qquad \rho := \partial_r p.$$

- $\Delta$: hedge ratio (shares in the replicating strategy).
- $\Gamma$: curvature; drives rebalancing intensity (liquidity/transaction-cost sensitivity).
- $\Theta$: time decay of the option value.
- Vega: sensitivity to volatility misspecification (positive for calls).
- $\rho$: sensitivity to the interest rate level.

*Remark* 0.4. In practice, "model risk" is precisely the sensitivity of prices to parameters: the Greeks are first-order diagnostics of how wrong-model inputs distort valuation.

## Part C — Change of probability measure and Girsanov: make the drift become $r$

### C.0 Why change the probability measure? (conceptual motivation)

Up to now, pricing was obtained via *hedging*, leading to the Black–Scholes PDE. We now explain why the same price can be written as a discounted expectation under a suitable probability measure.

The key idea is the following:

> *We may reweight scenarios so that the discounted stock price has zero drift.*

This reweighting is precisely what a *change of probability measure* does.

**Change of measure (basic definition).** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Given a nonnegative random variable $Z$ such that $\mathbb{E}[Z] = 1$, define a new probability measure $\mathbb{Q}$ by

$$\mathbb{Q}(A) := \mathbb{E}[\mathbf{1}_A Z], \qquad A \in \mathcal{F}.$$

We write

$$Z = \frac{d\mathbb{Q}}{d\mathbb{P}} \quad \text{(Radon–Nikodym derivative)}.$$

**Key identity.** For any integrable random variable $X$,

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}[XZ].$$

If $(Z_t)$ is a positive martingale with $\mathbb{E}[Z_T] = 1$ and $\frac{d\mathbb{Q}}{d\mathbb{P}}\big|_{\mathcal{F}_T} = Z_T$, then for $0 \le s \le t \le T$,

$$\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_s] = \frac{1}{Z_s}\,\mathbb{E}[Y Z_t \mid \mathcal{F}_s].$$

**On the board (intuition).**

- Changing measure = reweighting paths.
- Large $Z(\omega)$ makes scenario $\omega$ more likely under $\mathbb{Q}$.
- We will choose $Z$ so that the drift of the discounted stock disappears.

## C.1 Discounted stock and the drift we want to remove

Recall the Black–Scholes model under $\mathbb{P}$:

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t, \qquad B_t = e^{rt}, \qquad D_t = e^{-rt}.$$

Define the discounted stock

$$\bar{S}_t := D_t S_t = e^{-rt} S_t.$$

Since $D$ has finite variation, the product rule gives

$$d\bar{S}_t = e^{-rt} dS_t - r e^{-rt} S_t\, dt = (\mu - r)\bar{S}_t\, dt + \sigma \bar{S}_t\, dW_t.$$

Introduce the *market price of risk*

$$\theta := \frac{\mu - r}{\sigma}.$$

Then

$$d\bar{S}_t = \sigma \bar{S}_t(\theta\, dt + dW_t).$$

**Key observation.**

- Under $\mathbb{P}$, the discounted stock has drift $(\mu - r)$.
- We want a measure under which this drift vanishes.

## C.2 Exponential martingale and change of measure

Define the exponential process

$$Z_t := \exp\Big( - \theta W_t - \tfrac{1}{2}\theta^2 t\Big).$$

Since $\theta$ is constant, $(Z_t)$ is a true martingale and $\mathbb{E}[Z_T] = 1$.

Define a new probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = Z_T.$$

**Theorem 0.5** (Girsanov (constant drift case))**.** *Under $\mathbb{Q}$, the process*

$$\widetilde{W}_t := W_t + \theta t$$

*is a standard Brownian motion.*

**Interpretation.**

- Under $\mathbb{P}$: $W$ is Brownian.
- Under $\mathbb{Q}$: the Brownian motion is $\widetilde{W} = W + \theta t$.
- Drift has moved from the SDE into the measure.

## How to compute expectations under $\mathbb{Q}$ (examples)

Recall the constant-drift Girsanov setup:

$$Z_t = \exp\Big(-\theta W_t - \frac{1}{2}\theta^2 t\Big), \qquad \frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = Z_T, \qquad \widetilde{W}_t := W_t + \theta t \text{ is Brownian under } \mathbb{Q}.$$

*Remark* 0.6 (Change-of-measure identities). For any integrable $\mathcal{F}_t$-measurable random variable $Y$,

$$\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[Y\, Z_t].$$

More generally, for $0 \le s \le t \le T$ and integrable $\mathcal{F}_t$-measurable $Y$,

$$\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_s] = \frac{1}{Z_s}\,\mathbb{E}^{\mathbb{P}}[Y\, Z_t \mid \mathcal{F}_s].$$

**Example 1: distribution of $W_t$ under $\mathbb{Q}$**

Since $\widetilde{W}_t = W_t + \theta t$ is $\mathcal{N}(0,t)$ under $\mathbb{Q}$, we have

$$W_t = \widetilde{W}_t - \theta t \sim \mathcal{N}(-\theta t,\, t) \quad \text{under } \mathbb{Q}.$$

Hence for any bounded measurable $\varphi : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}^{\mathbb{Q}}[\varphi(W_t)] = \mathbb{E}\Big[\varphi(\widetilde{W}_t - \theta t)\Big] = \int_{\mathbb{R}} \varphi(x)\, \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x+\theta t)^2}{2t}\right) dx.$$

In particular,

$$\mathbb{E}^{\mathbb{Q}}[W_t] = -\theta t, \qquad \mathrm{Var}^{\mathbb{Q}}(W_t) = t.$$

**Example 2: exponential moments (sanity check)**

Let $a \in \mathbb{R}$. Using the Gaussian law under $\mathbb{Q}$:

$$\mathbb{E}^{\mathbb{Q}}\Big[e^{aW_t}\Big] = \exp\Big(a(-\theta t) + \frac{1}{2}a^2 t\Big) = \exp\Big(\frac{1}{2}a^2 t - a\theta t\Big).$$

Equivalently (same computation via $\mathbb{P}$),

$$\mathbb{E}^{\mathbb{Q}}\Big[e^{aW_t}\Big] = \mathbb{E}^{\mathbb{P}}\Big[e^{aW_t} Z_t\Big] = \mathbb{E}^{\mathbb{P}}\left[\exp\Big((a-\theta)W_t - \frac{1}{2}\theta^2 t\Big)\right] = \exp\Big(\frac{1}{2}(a-\theta)^2 t - \frac{1}{2}\theta^2 t\Big).$$

**Example 3: geometric Brownian motion and $\mathbb{E}^{\mathbb{Q}}[f(S_t)]$**

Consider a stock of the form

$$S_t = S_0 \exp(\alpha t + \sigma W_t), \qquad \sigma > 0.$$

Under $\mathbb{Q}$, since $W_t = \widetilde{W}_t - \theta t$, we can rewrite

$$S_t = S_0 \exp\Big((\alpha - \sigma\theta)t + \sigma\widetilde{W}_t\Big).$$

Hence, under $\mathbb{Q}$,

$$\ln S_t \sim \mathcal{N}\Big(\ln S_0 + (\alpha - \sigma\theta)t,\ \sigma^2 t\Big).$$

Therefore for any measurable $f$ such that $f(S_t)$ is integrable,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[f(S_t)] = \mathbb{E}\Big[f\Big(S_0 \exp((\alpha - \sigma\theta)t + \sigma\sqrt{t}\, Z)\Big)\Big], \qquad Z \sim \mathcal{N}(0,1).}$$

Equivalently, in integral form,

$$\mathbb{E}^{\mathbb{Q}}[f(S_t)] = \int_{\mathbb{R}} f\Big(S_0 e^{(\alpha-\sigma\theta)t+\sigma x}\Big)\, \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}\, dx.$$

**Example 4 (Black–Scholes drift switch): make the drift become $r$**

Suppose under $\mathbb{P}$,

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t, \qquad \mu \in \mathbb{R},\ \sigma > 0.$$

Fix a constant $r$ and choose

$$\theta := \frac{\mu - r}{\sigma}.$$

Then under $\mathbb{Q}$ (since $dW_t = d\widetilde{W}_t - \theta\, dt$),

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma(d\widetilde{W}_t - \theta\, dt) = r\, dt + \sigma\, d\widetilde{W}_t.$$

So

$$S_t = S_0 \exp\left(\Big(r - \frac{1}{2}\sigma^2\Big)t + \sigma\widetilde{W}_t\right) \quad \text{under } \mathbb{Q}.$$

Thus for any integrable $f$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[f(S_t)] = \mathbb{E}\Big[f\Big(S_0 \exp((r - \tfrac{1}{2}\sigma^2)t + \sigma\sqrt{t}\, Z)\Big)\Big], \qquad Z \sim \mathcal{N}(0,1).}$$

**Example 5: conditional expectation (useful for pricing)**

Under $\mathbb{Q}$, $\widetilde{W}_t - \widetilde{W}_s \sim \mathcal{N}(0, t-s)$ independent of $\mathcal{F}_s$. Hence, for $s < t$,

$$S_t = S_s \exp\left(\Big(r - \frac{1}{2}\sigma^2\Big)(t-s) + \sigma(\widetilde{W}_t - \widetilde{W}_s)\right).$$

Therefore, for any measurable $f$ with integrable $f(S_t)$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[f(S_t) \mid \mathcal{F}_s] = \Psi_{t-s}(S_s), \quad \Psi_u(x) := \mathbb{E}\Big[f\big(xe^{(r-\frac{1}{2}\sigma^2)u+\sigma\sqrt{u}\, Z}\big)\Big].}$$

This is the basic "lognormal transition" identity used in Black–Scholes computations.

### C.3 Apply to the discounted stock

Rewrite $dW_t = d\widetilde{W}_t - \theta\, dt$ and substitute into

$$d\bar{S}_t = \sigma\bar{S}_t(\theta\, dt + dW_t).$$

We obtain

$$d\bar{S}_t = \sigma\bar{S}_t\, d\widetilde{W}_t.$$

$$\boxed{(e^{-rt}S_t)_{t\le T} \text{ is a martingale under } \mathbb{Q}.}$$

Equivalently, under $\mathbb{Q}$ the stock price satisfies

$$\boxed{\frac{dS_t}{S_t} = r\,dt + \sigma\,d\widetilde{W}_t.}$$

**Punchline.**

- Drift changes: $\mu \longrightarrow r$.
- Volatility unchanged.
- Discounted prices become martingales.

## C.4 Link with pricing

Since $e^{-rt}S_t$ is a martingale under $\mathbb{Q}$, it is natural to expect that discounted derivative prices should also be martingales. This leads to the representation

$$p(t,s) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-r(T-t)}g(S_T) \mid S_t = s\Big],$$

which coincides with the solution of the Black–Scholes PDE.

*Remark* 0.7. Today this is presented as motivation. A full justification via no-arbitrage and self-financing strategies will be given in the next lecture.